Lens-Associated Core Collection from a Phase-Opposed Pearcey Composition

Auto BeamScientist Project

Abstract

A thin lens can change the fraction of optical power collected in a small on-axis region, but that contrast alone does not show that a composed input is responsible for the change. We numerically test a registered comparison for an apodized Pearcey source coherently combined with an equal-weight copy rotated by 90° and shifted in relative phase by π. The same normalized composed field is propagated with and without an ideal 0.060-m lens, and fixed-axis core efficiency is measured over $z = 0.04$–0.08 m. At the executed 192×192 sampling, the median efficiencies are 0.93463 with the lens and 0.02298 without it, a difference of 0.91165. A 384×384 confirmation gives a difference of 0.91405; an independent Fresnel calculation retains the positive sign. The zero-phase control also has a positive lens-minus-free-space difference, 0.78323. The registered directional prediction is therefore supported for these conditions, while the contrast is evidence for lens-associated collection, not for a composition-specific advantage or phase mechanism. Both reported phase-π noise checks fail, so robustness to those perturbations is not established.

1 Introduction

The fraction of optical power collected by a finite detector or aperture is distinct from peak intensity and total transmitted power. It depends on both the transverse intensity distribution and its position relative to the collection region. These distinctions matter when evaluating structured beams: a high local maximum need not imply high collected power, and a feature displaced from the collection axis can reduce fixed-axis collection.

Finite-energy Airy beams and Pearcey fields provide structured inputs whose propagation and focusing behavior have received substantial attention [3, 5, 4]. Pearcey-type integrals also belong to the wider study of optical catastrophe fields [6]. Here the source is a finite, Gaussian-apodized Pearcey field, not an ideal unbounded canonical beam. We ask a narrower numerical question about that source after coherent composition: does a thin lens produce a nonnegative change in its median fixed-axis core efficiency relative to free-space propagation?

The registered input combines the Pearcey field U with an equally weighted copy rotated by 90°, with relative phase π. This phase-opposed composition changes the sampled input field, but the tested comparison is between lens and free-space propagation of that same composition. It measures whether the lens-associated contrast persists for this particular input; it does not compare the composition against a matched single Pearcey source. A positive lens-minus-free-space difference therefore cannot by itself demonstrate that composition improves collection or that relative phase causes the lens response.

The numerical results support the registered direction: the difference in median fixed-axis core efficiency is positive in the primary calculation and retains its sign under mesh refinement and an independent paraxial propagator. The zero-phase control also gives a positive difference. We interpret these observations as a condition-specific lens-associated collection contrast. Composition specificity, phase dependence of that contrast, and the physical mechanism remain unresolved.

2 Related Work

Airy-beam literature addresses finite-energy realizations and the control and application of self-accelerating beams [3, 5]. Pearcey-beam studies discuss autofocusing and self-healing, while work on swallowtail and butterfly beams treats other members of the optical-catastrophe family [4, 6]. These sources provide relevant beam-family and propagation context. The available evidence for this manuscript consists of verified bibliographic metadata rather than full-text comparisons; it therefore does not support detailed claims about differences in field definitions, collection metrics, or focusing conditions, or an inference of a novelty gap.

A separate project report in the supplied portfolio studies rotational superposition and core-energy concentration for other constructions and objectives. Its relation to the present work is the use of coherent rotational composition and a finite-core energy metric. The question here differs: it tests the lens-minus-free-space contrast for one phase-opposed Pearcey composition, rather than whether a composition improves collection relative to a single input or a constrained optimization control. This distinction matters because the present arms change the lens, not the source composition.

The phase factor associated with a vortex and its relationship to Laguerre–Gaussian modes are established topics [2]; Gaussian resonator theory provides context for Gaussian-mode propagation [1]. Neither is the transformation tested here: the input is a composed apodized Pearcey field, with no vortex phase imprinted. These citations are included as verified bibliographic records, not as evidence that the present comparison is novel.

3 Theoretical model

3.1 Input field and coherent composition

Let x, y denote transverse coordinates and $r^2 = x^2 + y^2$. The executed source is a Pearcey-family field with Gaussian envelope,

$$V(x,y) = C_4(x/s, y/s)\exp(-r^2/w^2), \qquad C_4(u,v) = \int_{-\infty}^{\infty} \exp\big[i(t^4 + vt^2 + ut)\big]\, dt, \tag{1}$$

inside the physical aperture $r \le 0.90\,\mathrm{mm}$, and is zero outside it. The source scale is $s = 0.30\,\mathrm{mm}$, and the Gaussian envelope radius is $w = 0.30\,\mathrm{mm}$. These parameters have separate roles in the field definition, even though their executed values coincide. The primary source uses Pearcey order $n = 4$; butterfly order $n = 6$ and its controls are not used.

The source is normalized to input power P on the sampled grid:

$$U_0(x_j, y_k) = \sqrt{\frac{P}{\Delta x^2 \sum_{a,b} |V_{ab}|^2}}\, V(x_j, y_k). \tag{2}$$

Here $\Delta x = \Delta y$ is the grid spacing, and V includes the aperture restriction. For composition, define $\mathcal{R}_\theta U$ as the field rotated by angle θ about the physical grid origin. The normalized coherent pair is

$$U_c = \sqrt{\frac{P}{\int |U + ae^{i\delta}\mathcal{R}_\theta U|^2\, dA}}\, \left(U + ae^{i\delta}\mathcal{R}_\theta U\right). \tag{3}$$

The registered phase-opposed case has $a = 1$, $\theta = \pi/2$, and $\delta = \pi$, so its unnormalized field is $U - \mathcal{R}_{\pi/2}U$. The zero-phase control has the same weight and rotation but $\delta = 0$. Normalization is performed after forming the coherent sum; the terms are not separately assigned fixed fractions of the final power. The executed phase-opposed field is nonzero, so this normalization is defined. An exactly zero coherent field cannot be normalized, although other arms can still be measured.

Rotation is implemented by sampling the real and imaginary fields independently with bilinear interpolation, using constant-zero boundaries outside the physical grid around its origin. The rotated arm is normalized to input power before composition. This finite-grid operation can introduce interpolation effects; the mesh checks below assess stability of the reported comparison, not exact rotational invariance.

3.2 Lens and propagation

The ideal thin lens acts on the composed input through the quadratic phase

$$T_f(x, y) = \exp\left[-i\frac{\pi(x^2 + y^2)}{\lambda f}\right], \tag{4}$$

where $f = 0.060\,\mathrm{m}$ and $\lambda = 633\,\mathrm{nm}$. This phase is applied before propagation in the lens arm. The free-space arm propagates the identical composed input without this lens phase.

Propagation uses the angular-spectrum method (ASM):

$$U_z = \mathcal{F}^{-1}\left[\mathcal{F}(U_0)B_z \exp\left(i2\pi z\sqrt{\lambda^{-2} - f_x^2 - f_y^2}\right)\right]. \tag{5}$$

In the lens arm, U_0 in Eq. (5) denotes the normalized field after application of Eq. (4); in the free-space arm it denotes the normalized composed input. The transfer mask uses the conservative radial cutoff

$$\nu_c = \min\left[\lambda^{-1}, \frac{\lambda^{-1}}{\sqrt{1 + (2z/L)^2}}\right], \tag{6}$$

where L is the computational window width. This is a conservative radial bandwidth cutoff, not the full Matsushima passband construction. The numerical model is scalar and assumes ideal coherent propagation; it does not model lens aberrations, material losses, or experimental noise.

3.3 Primary observable

At each axial plane, the fixed-axis core efficiency is the fraction of input power inside a circle of radius r_c:

$$C(z) = P^{-1}\int_{x^2+y^2\le r_c^2} |U_z(x, y)|^2\, dA, \qquad \overline{C} = \mathrm{median}_{z\in\mathcal{Z}}\, C(z). \tag{7}$$

The primary core radius is $r_c = 0.100\,\mathrm{mm}$, and $\mathcal{Z}$ spans 0.040 to 0.080 m. The registered signed effect is

$$\Delta_{\mathrm{lens-free}} = \mathrm{median}_{z\in\mathcal{Z}}\, C_{\mathrm{lens}}(z) - \mathrm{median}_{z\in\mathcal{Z}}\, C_{\mathrm{free}}(z). \tag{8}$$

The hypothesis predicts $\Delta_{\mathrm{lens-free}} \ge 0$. The reported efficiencies are normalized to input power; they are not peak intensities or absolute collected powers. Diagnostic centroid-aligned values use a separate circle centered on each plane's intensity centroid and do not replace the fixed-axis primary statistic.

4 Numerical methods

The primary executed setup uses a 192×192 grid over a 2.4 mm square window, wavelength 633 nm, input power 1.0 W, Pearcey scale 0.30 mm, envelope radius 0.30 mm, and focal length 0.060 m. Coordinates follow

$$x_j = (j - \lfloor N/2 \rfloor)L/N, \qquad y_k = (k - \lfloor N/2 \rfloor)L/N.$$

The physical aperture is a disk of radius 0.90 mm. The registered axial window contains 11 planes from 0.040 to 0.080 m, separated by 4 mm. The lens and free-space arms share the same constructed input and power normalization.

The canonical integral in Eq. (1) is evaluated by contour rotation $t = q\exp[i\pi/(2n)]$, with q sampled uniformly on $[-3, 3]$ and trapezoidal quadrature using 192 nodes. Source convergence is evaluated with 384 quadrature nodes. The primary circular-core integral is computed as the sum of pixel intensities whose centers satisfy $r \le r_c$, multiplied by Δx^2. Thus the implemented core integral is an area-weighted pixel-center quadrature, rather than a fractional-pixel boundary integration.

The registered confirmation includes a 384×384 calculation, an expanded-window calculation, denser axial sampling, a held-out wavelength, phase quantization, and an independent paraxial Fresnel propagation comparison. Confirmation checks for numerical validity, reference physics, sampling, and independent-observable validation passed for the phase-opposed condition. The two reported phase-π noise probes, labeled `noise_107` and `noise_109`, did not pass their robustness checks. The zero-phase control passed its listed robustness checks, but that result does not validate robustness of the distinct phase-π input. Repeated phase-π registrations use identical source and composition parameters and are numerical reproducibility entries, not independent physical or stochastic replications.

5 Results and physical interpretation

5.1 Registered lens-minus-free-space comparison

The executed primary calculation gives median fixed-axis core efficiencies of 0.9346275 for the lens arm and 0.0229762 for the free-space arm. Their difference is

$$\Delta_{\text{lens}-\text{free}} = 0.9116513,$$

which supports the registered nonnegative direction. The confirmation summary for the repeated phase-opposed entries reports a 384×384 ASM difference of 0.9140488, from medians 0.9374980 and 0.0234492. Refinement therefore preserves the sign and changes the magnitude by about 0.0024 relative to the primary calculation. The independent Fresnel calculation gives 0.9140897, with medians 0.9375389 and 0.0234492. These are numerical evaluations of the same registered construction, not independent experimental observations.

The positive difference is large because the lens and free-space arms have very different core efficiencies for this composed input. It is accurately described as a lens-associated increase in the normalized fraction collected by the fixed on-axis core over the sampled axial interval. It should not be interpreted as an absolute concentration gain independent of the chosen core, as a change in delivered power, or as evidence that composition is responsible for the lens response. The comparison does not contain the matched lensed single-source arm required to measure a composition-specific contrast.

Table 1: Registered lens-minus-free-space core-efficiency comparisons. Each entry is the difference of the two arm-wise axial medians, not the median of pointwise differences.

Construction and calculation	Propagator	Lens median	Free-space median	Difference
Phase π, primary 192^2	ASM	0.93463	0.02298	0.91165
Phase π, confirmation 384^2	ASM	0.93750	0.02345	0.91405
Phase π, confirmation 384^2	Fresnel	0.93754	0.02345	0.91409
Zero phase, primary 192^2	ASM	0.99000	0.20677	0.78323

5.2 Zero-phase control and composition specificity

The zero-phase quarter-turn control also has a positive lens-minus-free-space difference:

$$0.9900003 - 0.2067746 = 0.7832257.$$

Its lensed median is higher than the phase-π lensed median, while its free-space median is also much higher. The positive contrast is therefore not unique to the phase-opposed construction. Comparing these two lens-minus-free-space differences does not isolate a phase effect, because the compositions are distinct inputs and no matched causal estimate of a phase-by-lens interaction was preregistered.

Available diagnostics likewise do not establish composition-specific benefit. In the phase-π condition, the lensed single-source arm has a reported core efficiency of 0.98924, compared with 0.93463 for the coherent composed arm; the composed input is not shown to outperform a single-source input under the lens. The phase-π coherent field has a sampled single-field overlap of 0.117, but a field overlap alone does not establish improved collection or identify a mechanism. The zero-phase construction has a reported overlap of 0.99313 with the single-source reference. This is consistent with little difference in that particular sampled comparison, but is not a broad equivalence test.

A four-parameter classical control was optimized for the frozen lens-versus-free-space comparison and reached a reported effect of 0.82550. It is not a matched composition-versus-control test, so it cannot establish superiority or infer a mechanism. Its value does, however, reinforce that a large lens-minus-free-space contrast is not in itself distinctive evidence for the rotational composition.

5.3 Axial behavior and interpretation

The primary metric summarizes fixed-axis collection at the registered planes rather than selecting the best plane. The phase-π lensed axial curve rises from 0.60335 at $z = 0.040\,\mathrm{m}$, reaches values close to unity near $z = 0.056$–$0.060\,\mathrm{m}$, and declines to 0.39311 at $z = 0.080\,\mathrm{m}$. In free space, its core fraction remains close to 0.023 throughout the interval. This contrast is consistent with focusing by the ideal lens into the chosen on-axis core over part of the propagation range, followed by a decrease in core collection beyond the near-focus region. It does not imply monotonic increase over the full axial interval.

The single-source lensed diagnostic has median fixed-axis core efficiency 0.98924, while its centroid-aligned value is 0.99149. For the phase-π coherent lensed arm, the corresponding values are 0.93463 and 0.93278. These alignment diagnostics show that fixed-axis and recentered collection are not interchangeable, but do not decompose the contrast into a causal displacement contribution. The phase-opposed lensed arm also has a reported sidelobe fraction of 0.06537 and a brightest-pixel excursion of about $146\,\mu\mathrm{m}$. These diagnostics qualify a core-only interpretation without changing the registered endpoint.

5.4 Scope and limitations

The result applies to the specified scalar Pearcey source, aperture, equal-weight quarter-turn composition, phase π, ideal lens, fixed $0.100\,\mathrm{mm}$ core, and 0.040–$0.080\,\mathrm{m}$ interval. The area integral is evaluated with pixel-center inclusion, and rotation uses bilinear interpolation on a finite grid. Mesh, window, axial-sampling, and independent-propagator checks support the positive sign for the registered comparison. Both phase-π noise checks fail, however, so robustness to those tested perturbations is not established. The successful zero-phase robustness checks cannot be transferred to the phase-opposed construction.

Most importantly, the existing comparison tests lens versus free-space propagation for a composed input. It does not test whether phase-opposed composition adds value relative to a matched single Pearcey input under the same lens, nor whether the lens response differs between composed and single-source fields. Such claims require direct matched arms; a non-saturated endpoint may also be more discriminating than a lensed fixed-core value near unity. The result is a supported numerical observation for a frozen condition, not a confirmed composition mechanism, an experimental result, or a claim of novelty.

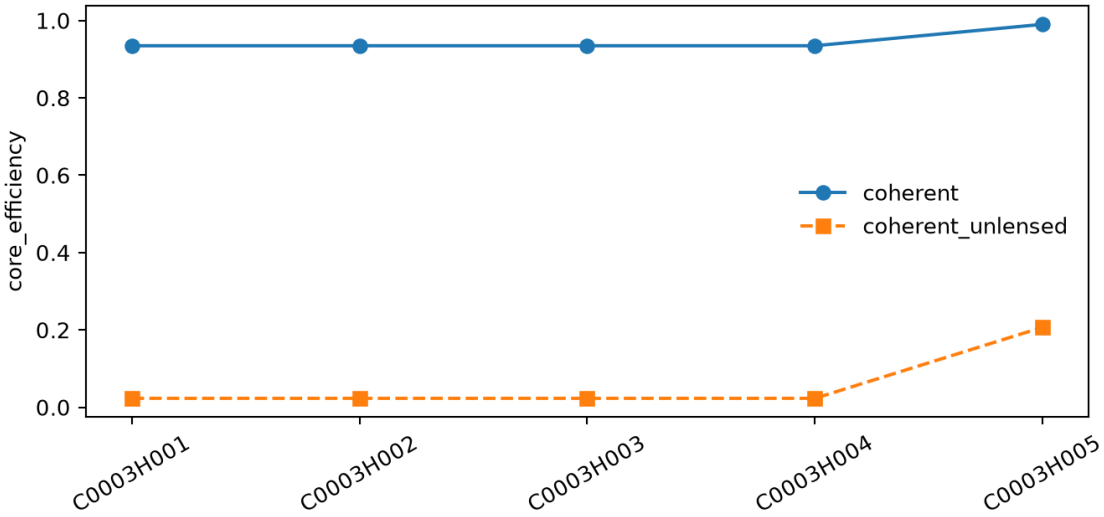
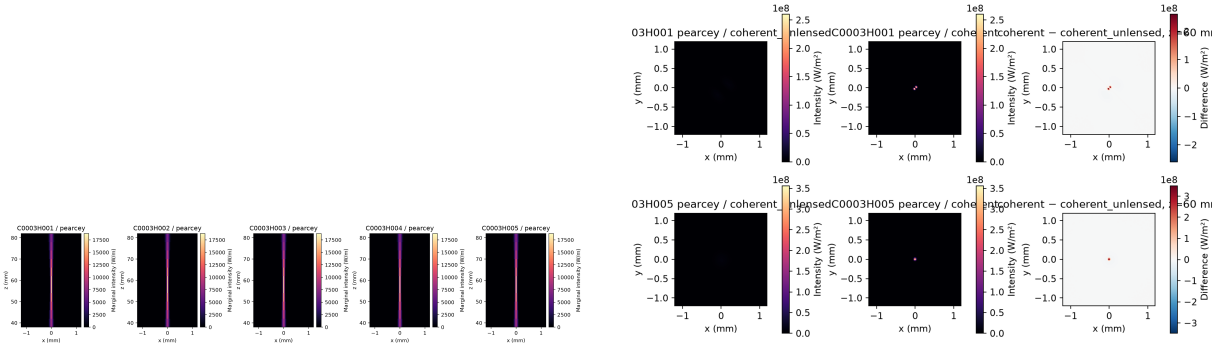


Figure 1: Overview of the recorded numerical results. Quantitative claims and the signed effect are taken from the executed numerical records and Table 1.



(a) Recorded propagation view.

(b) Recorded comparison diagnostics.

Figure 2: Complementary views of propagation and comparison diagnostics. The figures are qualitative guides; they do not replace the axial medians or establish a composition-specific mechanism.

6 Conclusions

For the registered equal-weight, 90°-rotated Pearcey pair at relative phase π, the median fixed-axis core-efficiency difference between ideal-lens and free-space propagation is positive: 0.91165 in the primary calculation and 0.91405 in the 384×384 ASM confirmation. An independent Fresnel calculation gives 0.91409. The zero-phase control also gives a positive lens-associated difference, 0.78323. These observations support the registered directional prediction for the specified construction and numerical conditions.

The result establishes neither a composition-specific collection advantage nor a phase-specific mechanism. The lensed phase-π composition is not shown to exceed the available single-source lensed diagnostic, and the failed phase-π noise checks limit robustness claims. A direct, power-matched lensed comparison among the phase-opposed pair, zero-phase pair, and single Pearcey source is needed to determine whether composition contributes beyond ordinary lens action.

Acknowledgement

Auto BeamScientist adapts Sakana AI Scientist-v2 [7].

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